

Mathematical Reflections — S702

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Prove that for each positive integer c there are infinitely many positive integers n such that $n! + c$ is not the square of an integer.

Solution

Our proof naturally splits into two cases: when c itself is a square and when c is not a square.

Let's look at when c is not a square. It suffices to show that there exists a prime p such that $v_p(n! + c)$ is odd, because if $n! + c$ were a square then it is of the form $p_1^{2e_1} \cdot p_2^{2e_2} \cdots p_n^{2e_n}$, for some primes (p_n) and some positive integers (e_n) . Since c isn't a square, there exists some prime p and a positive integer k such that $v_p(c) = 2k - 1$. Let $n \geq 2c \geq 2p^{2k-1}$ such that $n!$ will contain the factors p^{2k-1} and $2p^{2k-1}$ and so $p^{2k} \mid n!$ whilst $p^{2k} \nmid c$. So that means that $p^{2k-1} \mid [n! + c]$ but $p^{2k} \nmid [n! + c]$ and thus $v_p(n! + c) = 2k - 1$ which is odd and this shows that it isn't square, for infinitely many $n \geq 2c$.

Now what about when c is a square. Let $c = m^2$ for some positive integer m . We claim that if $v_2(n!)$ is odd then $n! + m^2$ is not a square. Assume, for the sake of contradiction, that $v_2(n!)$ is odd and that $n! + m^2 = a^2$ for some positive integer a . We have $n! = (a - m)(a + m)$. Now suppose that $v_2(a) = a'$ and $v_2(m) = m'$, now for some odd integers λ_a, λ_m we can write $a = 2^{a'} \lambda_a$ and $m = 2^{m'} \lambda_m$. Now notice that $v_2(a + m) = v_2(a - m) = \min(a', m')$ and so $v_2((a + m)(a - m)) = 2 \min(a', m') = v_2(n!)$ but this is a contradiction since the left hand side is even and the right hand side is odd. Now we just need to show that there are infinitely many n such that $v_2(n!)$ is odd. Notice that $v_2((n + 1)!) = v_2(n!) + v_2(n + 1)$, and so if $v_2(n + 1)$ is odd the parity of $v_2((n + 1)!)$ and $v_2(n!)$ are distinct and then by pigeonhole principle one of them must be odd, finally there are infinitely many n such that $v_2(n + 1)$ is odd because any number of the form 2^{2k-1} for positive integer k will have an odd 2-adic valuation. ■

Remark: It seems as if this is essentially Alessandro Ventullo's solution just without the theory of quadratic residues (this is because I haven't learnt QR yet..). See Mathematical Reflections Issue 3 2025.